# GlucoTwin

A two-hour warning, and a what-if sandbox, for type 2 diabetes.

six seven · [COLLEGE] · [INCUBATOR]

# India does not have a diabetes problem. It has a daily one.

- 101 million people with diabetes in India. 136 million with prediabetes. [CITE]
- Most of them see a doctor two or three times a year.
- The disease acts every day, between those visits.

ICMR-INDIAB, 2023. Citation to be completed before submission.

# The dangerous hours are the ones nobody sees.

- Glucose above 180 mg/dL for 30 minutes does damage quietly.
- The worst window is usually after the evening meal.
- A 09:00 fasting reading cannot show it.
- The next HbA1c test arrives three months later.

# So we rebuilt the patient as a twin.

- Static record: labs, medications, genotype, history.
- Dynamic stream: 14 days of CGM, heart rate, HRV, steps, sleep.
- Output: two-hour event risk, with a calibrated probability.
- Output: a what-if sandbox that runs on this patient's physiology.

# 1,500 synthetic patients. 14 days each.

**6.05 M**

glucose readings

**5 min**

sampling interval

**55**

model features

**1 CPU**

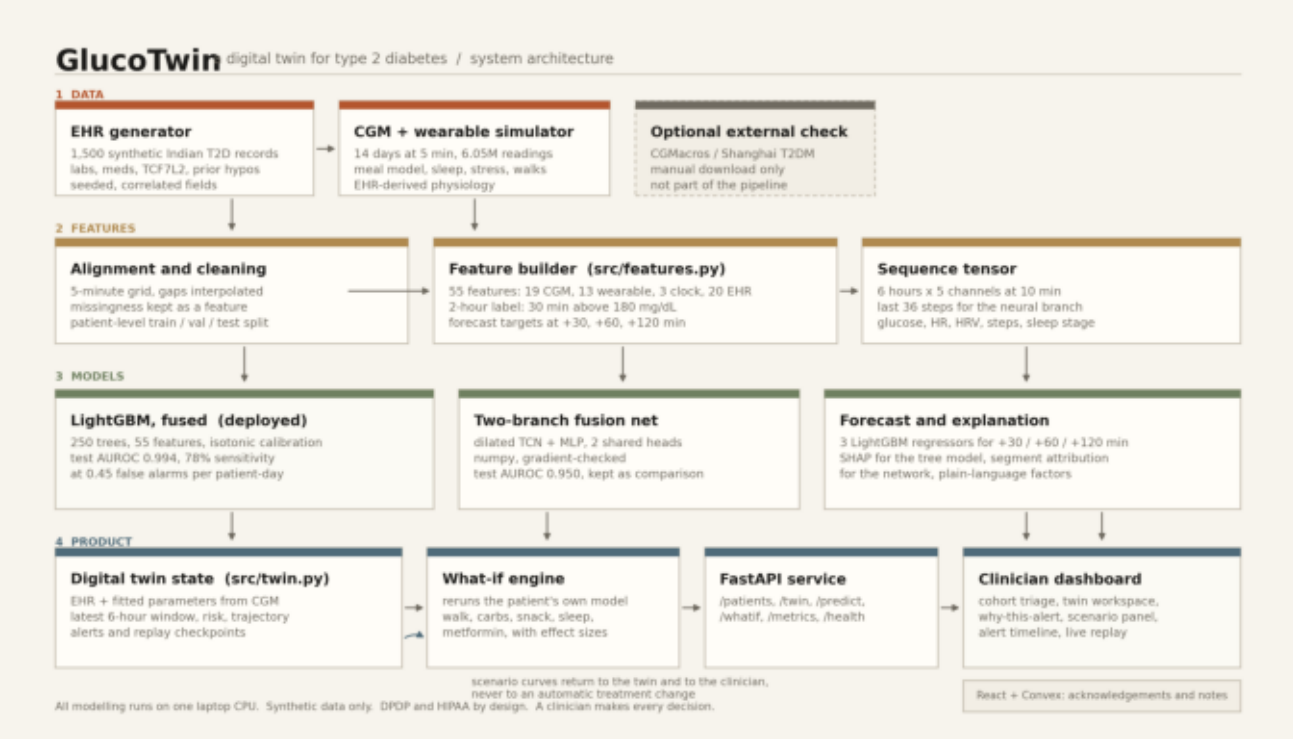
to train and run

Seed 42. The whole cohort is regenerated by one command.

# Synthetic is a design choice, not a shortcut.

- No real patient data, so no consent or re-identification risk.
- The record drives the physiology, so the two streams are genuinely linked.
- Fasting glucose sets the baseline. BMI, duration and genotype set sensitivity.
- The cost: results do not transfer to real patients without validation.

# One pipeline, from a synthetic record to a clinical screen.




Data generation, features, models, twin, what-if, API, dashboard. External validation is optional and off to the side.

# The label is a real clinical event.

1 if glucose stays above 180 mg/dL for 30 minutes inside the next 2 hours

- 41.1% of test windows are positive.
- The cohort is deliberately uncontrolled, so events are not rare.
- Split by patient: 900 train, 300 validation, 300 test.

# 55 features. Four blocks. One fused vector.

- 19 CGM: level, means, variability, slopes, time above range, gaps.
- 13 wearable: heart rate, HRV, steps, sleep and deep sleep.
- 3 clock: time of day and night flag.
- 20 record: age, BMI, waist, HbA1c, eGFR, lipids, medications, TCF7L2.

Plus a 6-hour, 5-channel tensor for the neural branch.

# Five models. The tree won.

| Model | Input | AUROC | Sensitivity at 0.5 FA/day |
|---|---|---|---|
| Persistence | current glucose | 0.797 | 6% |
| LightGBM, CGM only | 15 CGM features | 0.984 | 70% |
| LightGBM, EHR only | 20 record fields | 0.791 | 5% |
| **LightGBM, fused** | **all 55 features** | **0.994** | **78%** |
| Fusion net | 6 h x 5 channels + record | 0.950 | 24% |

Test patients only. Calibrated probabilities. Threshold fixed on validation.

# The network was supposed to win. It did not.

- Two-branch TCN + MLP, written in numpy, gradient-checked.
- AUROC 0.9496, against 0.9941 for the tree model.
- Two epochs on one CPU core is not enough to beat 250 boosted trees.
- So the tree is deployed, and the network stays in the ablation.

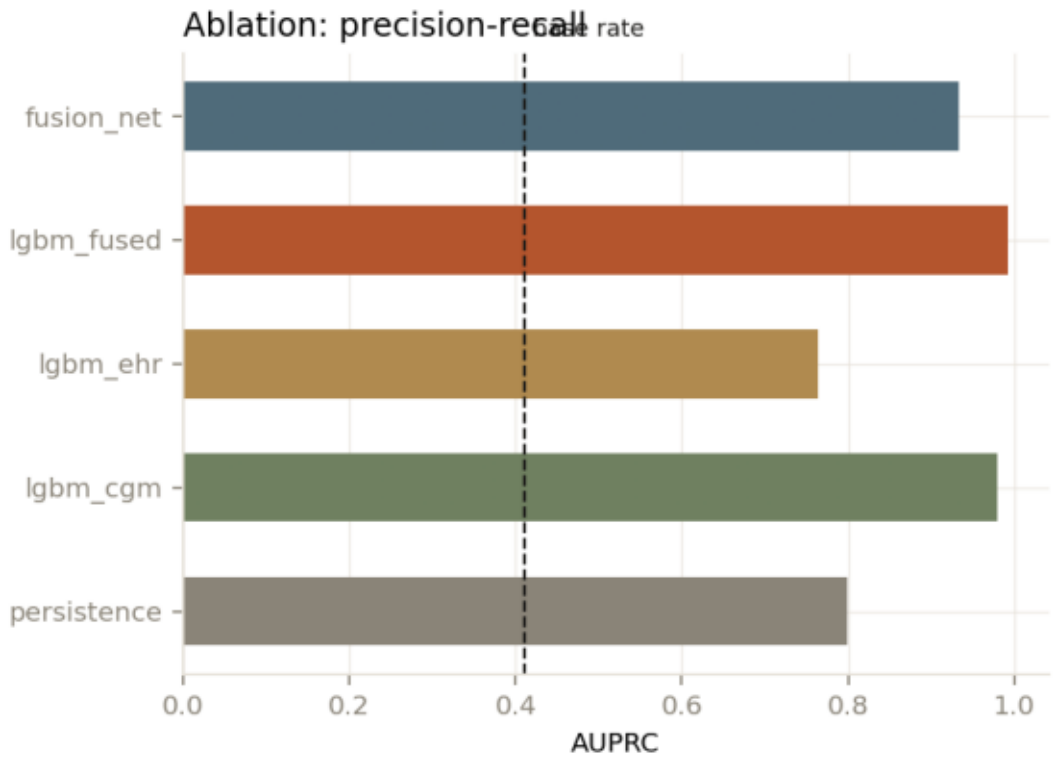
This is the comparison the brief asked for, reported as it is.

# Results on 300 patients the model never saw.

| 0.994 | 0.992 | 78% | 53 min |
| --- | --- | --- | --- |
| AUROC | AUPRC | sensitivity | mean warning |

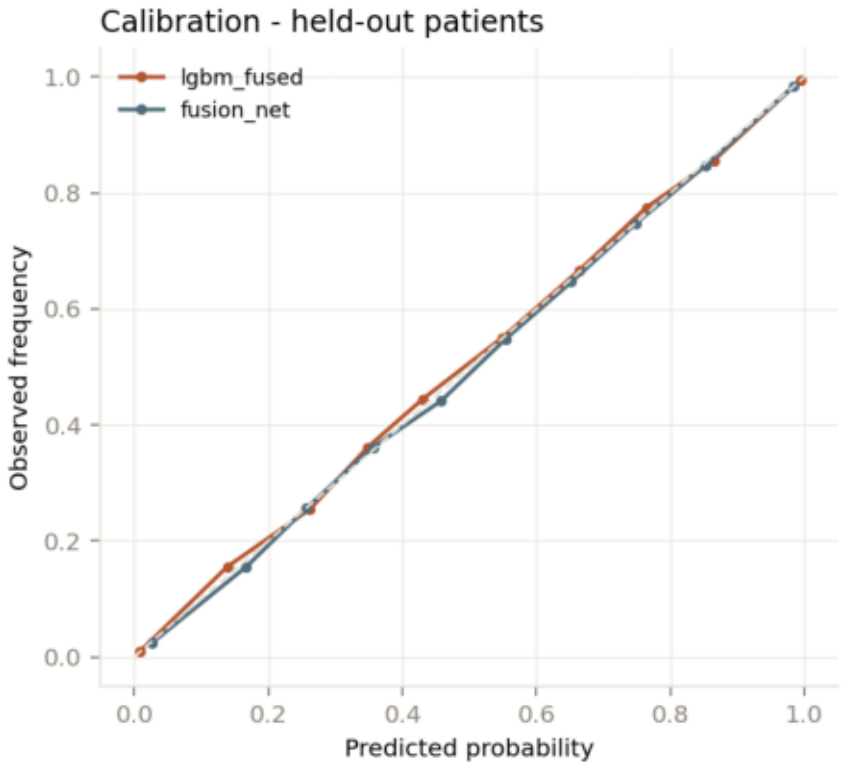
Brier 0.028. 0.45 false alarms per patient-day. Dense 15-minute run: 81% sensitivity, 60 min mean lead.

# Every input helps. The sensor helps most.



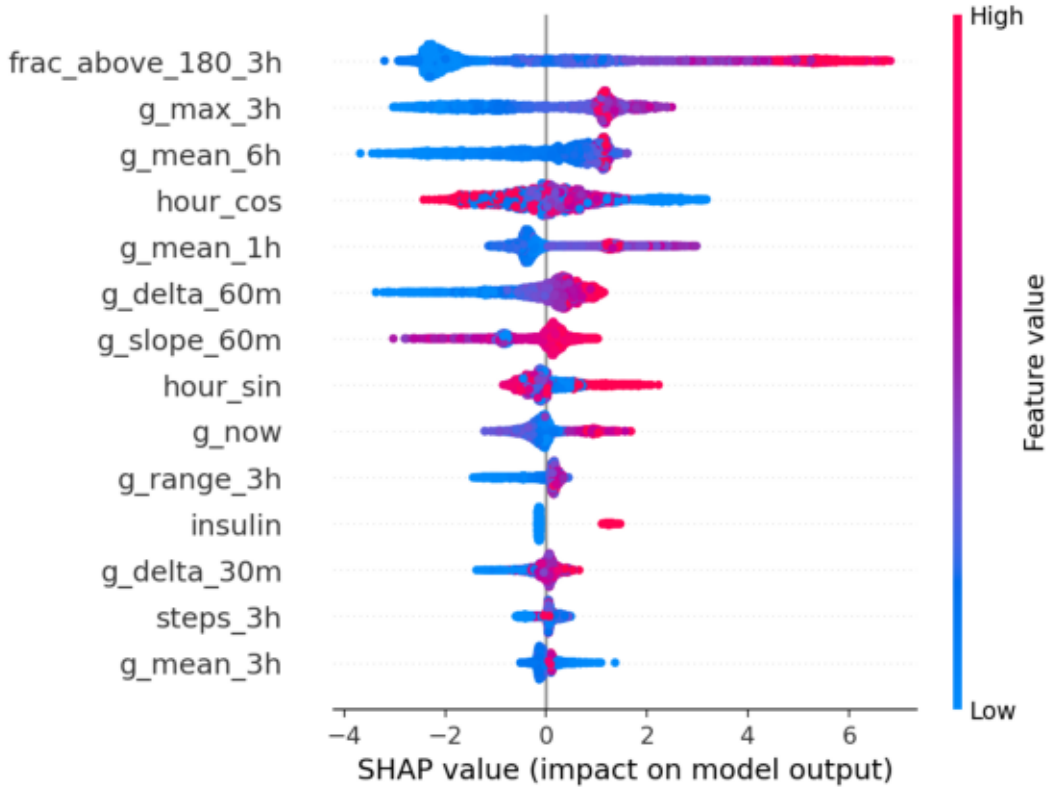

Same checkpoints for every model. EHR-only cannot rank moments inside one patient's day, so its within-patient AUROC is 0.5.

# A probability you can act on, not just a ranking.




Brier 0.028 after isotonic calibration. Forecast RMSE 11 / 14 / 23 mg/dL at 30 / 60 / 120 minutes.

# Why this alert, in words a clinician can use.

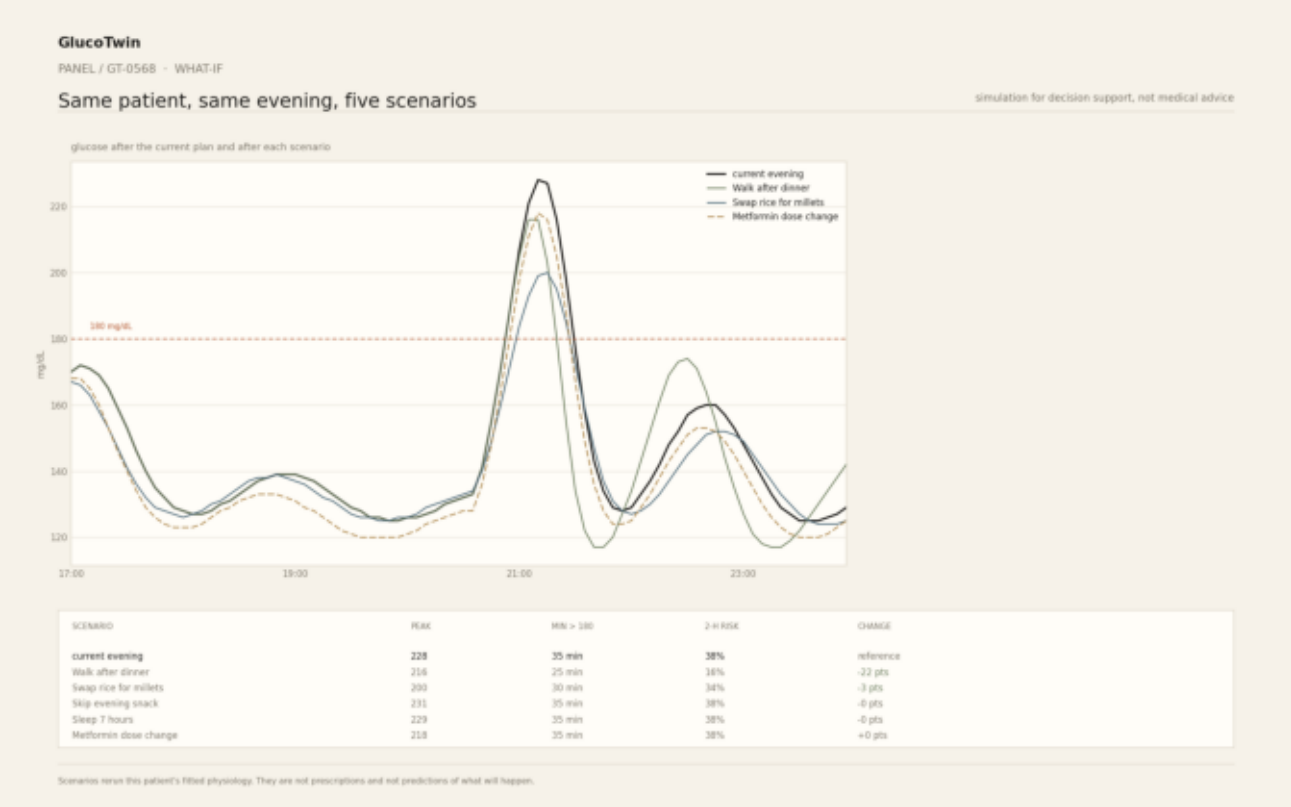



Recent time above range and the size of the current excursion dominate. The record explains who is at risk, not when.

# The twin fits the patient, not the population.

- Insulin sensitivity and basal glucose are fitted from the CGM signal.
- Held-out check: R2 0.82, median error 8.4%, Spearman 0.91.
- Those numbers validate the fitting procedure, not the physiology.
- The twin updates as new readings arrive, and keeps its own risk trace.

# A walk after dinner cuts the peak by 12 mg/dL.




Same patient, same evening. Risk falls from 38% to 16%. Swapping rice for millets cuts the peak by 28 mg/dL. Skipping the snack does almost nothing.

# Built for a clinic, not for a demo.







Cohort sorted by two-hour risk. Twin workspace with forecast band, gauge, time in range, wearables and the alert timeline.

# What we did not prove.

- This is synthetic data. No clinical accuracy is claimed.
- External validation is written but not run.
- About 2.7 alerts per patient-day at the deployed threshold.
- Subgroup fairness is not measured.
- It is not a medical device, and it is not approved.

DPDP Act and HIPAA by design. Human in the loop. No automatic treatment change.

# What comes next.

- A prospective 8 to 12 week study in one clinic, threshold frozen in advance.
- ABDM and ABHA records, Apple Health and Google Fit, CGM vendor APIs.
- Federated learning across clinics, so records never move.
- A fairer cohort: CKD, SGLT2 and GLP-1, and well-controlled patients.

# Thank you.

GlucoTwin is a decision aid, not a doctor. The clinician decides.

six seven · [EMAIL] · Demo video: [LINK] · Code: [REPOSITORY]